CUSTOMER SHOPPING BEHAVIOR ANALYSIS

End-to-end data analytics portfolio report

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Tools & technologies
Python · Pandas · NumPy · Matplotlib · Seaborn · SQL · Power BI

PROJECT PURPOSE

Analyze customer purchase records to describe spending patterns, product-category performance, customer characteristics, discount usage, subscription status, and shopping preferences. Translate the observed patterns into practical questions and recommendations for a retail business.

DATA PROVENANCE & TRANSPARENCY

This report is a newly formatted analysis based on the CSV dataset supplied with the reference project materials. The supplied reference report/tutorial included WeXllent branding and a YouTube link; those promotional elements are not reproduced here. The reference materials informed the initial analysis scope. This report identifies the dataset-based calculations performed for this rebuild and does not claim that the source dataset or original tutorial was created by the author.

1. Executive Summary

The supplied dataset contains **3,900 purchase records** and **18 original columns**. The raw file contains **37 missing review ratings** (0.95% of records) and **0** fully duplicated rows. The analysis below uses category-level median imputation for missing ratings and derives age-quartile and approximate purchase-frequency features.

3,900	$233,081.00	$59.76	3.75/5
Purchase records	Total purchase amount	Average purchase amount	Average review rating*

*Average rating includes imputed values for the 37 missing ratings.

Key descriptive observations

• **Category:** Clothing has the highest total purchase amount in this dataset ($104,264.00).

• **Gender split:** Male records contribute the larger total purchase amount ($157,890.00); this is a descriptive comparison, not a causal explanation.

• **Payment:** PayPal is the most frequently recorded payment method (677 records).

• **Interpretation:** The file represents purchase records, not a dated event history. It supports cross-sectional comparisons but does not support month-over-month trends or causal claims.

2. Dataset & Data Quality

The dataset describes customer purchase records across product, demographic, payment, shipping, discount, and review fields.

Area	Fields / examples
Customer profile	Customer ID, age, gender, location, subscription status
Purchase details	Item purchased, category, purchase amount, season, size, color
Shopping behavior	Discount applied, promo code used, previous purchases, frequency of purchases
Service & feedback	Review rating, shipping type, payment method

Raw data quality checks

Check	Result
Rows / columns	3,900 rows / 18 columns
Missing review ratings	37 (0.95%)
Fully duplicated rows	0
Customer ID duplicates	0

Cleaning decisions

1. Impute missing Review Rating values using the median within each product category. This retains records while allowing category-specific rating distributions to influence the replacement value.

2. Standardize column names to lowercase snake_case and rename purchase_amount_(usd) to purchase_amount.

3. Remove promo_code_used only after confirming it is identical to discount_applied across every row in this dataset. This is a dataset-specific redundancy check, not a general rule.

4. Create age_group using four quantile-based groups. These labels represent dataset quartiles, not fixed demographic age bands.

5. Map purchase frequency labels to approximate day values (for example, Weekly = 7 and Monthly = 30). These are representative intervals, not observed transaction gaps.

3. Analytical Methodology

Python workflow

Load the CSV with Pandas; inspect dimensions, data types, summary statistics, nulls, duplicates, and unique values; clean missing ratings and column names; engineer age and purchase-frequency features; then export a cleaned CSV for downstream analysis.

SQL workflow

The supplied project scope identifies ten business questions: revenue by gender; high-spending discount users; top-rated products; shipping comparison; subscribers versus non-subscribers; discount-dependent products; customer segmentation; top products by category; repeat buyers and subscriptions; and revenue by age group.

Power BI workflow

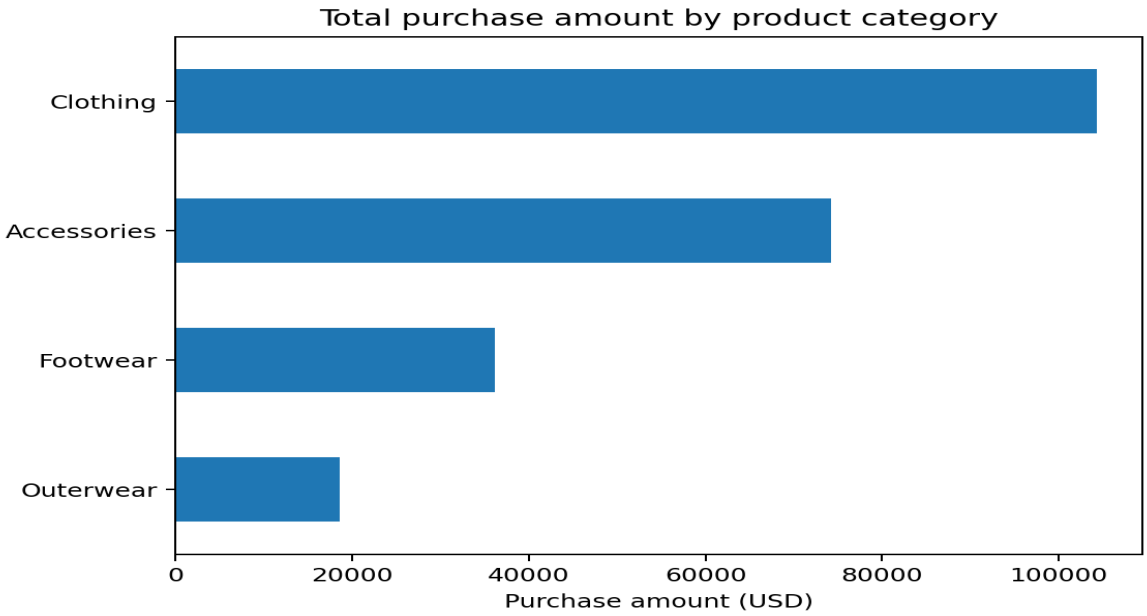
The reference dashboard design includes customer and revenue KPIs, subscription and gender distribution, category revenue, revenue by age group, top products by revenue, and slicers for subscription, gender, category, shipping type, and payment method. The dashboard file itself should be rebuilt and validated separately before this report is treated as final dashboard evidence.

Analytical boundaries

The dataset contains purchase amounts but does not provide a transaction date, cost of goods, profit, customer acquisition source, or a unique transaction timestamp. Therefore, this report does not claim to calculate profit, customer lifetime value, conversion rate, or time-series growth.

4. Descriptive Findings

Purchase amount by product category

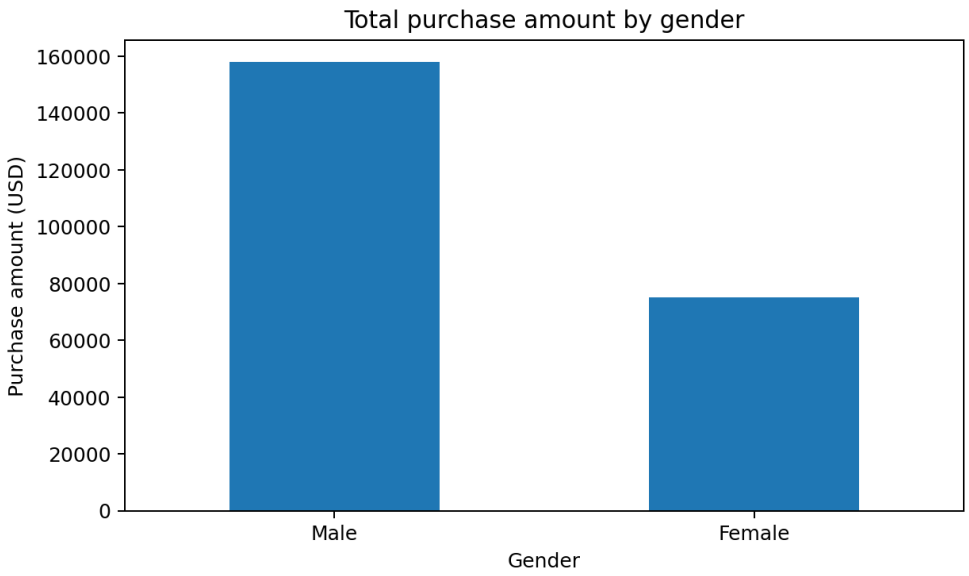


Category totals and averages

Category	Records	Total amount	Average amount
Clothing	1,737	$104,264.00	$60.03
Accessories	1,240	$74,200.00	$59.84
Footwear	599	$36,093.00	$60.26
Outerwear	324	$18,524.00	$57.17

5. Customer & Shopping Patterns

Purchase amount by gender



Subscription comparison

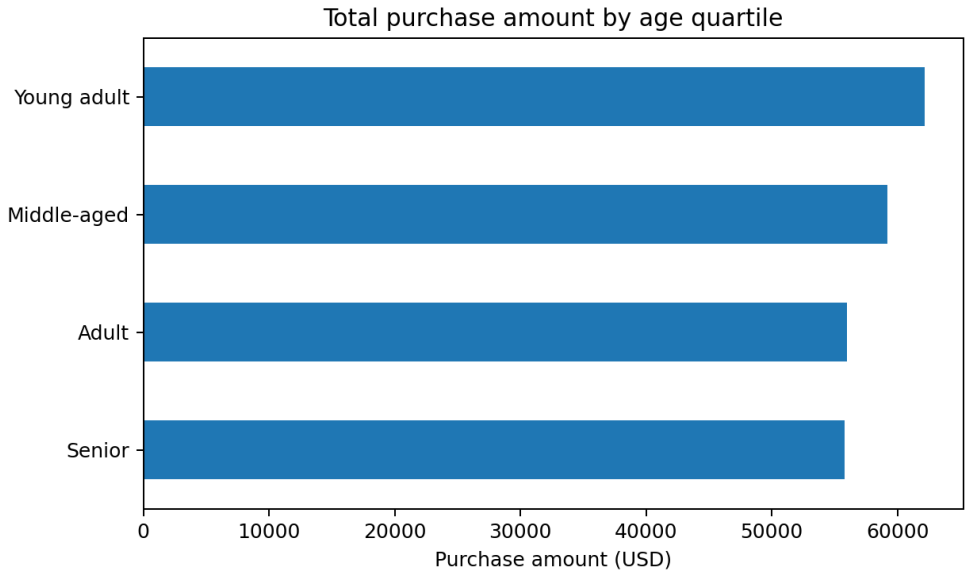
Subscription	Records	Total amount	Average amount
No	2,847	$170,436.00	$59.87
Yes	1,053	$62,645.00	$59.49

Discount comparison

Discount applied	Records	Total amount	Average amount
No	2,223	$133,670.00	$60.13
Yes	1,677	$99,411.00	$59.28

These comparisons describe the records grouped by the available fields. They do not establish that a subscription or discount caused a change in spending.

6. Age & Product Analysis



Revenue by age quartile

Age quartile	Total purchase amount
Young adult	$62,143.00
Middle-aged	$59,197.00
Adult	$55,978.00
Senior	$55,763.00

Top five items by total purchase amount

Item	Total purchase amount
Blouse	$10,410.00
Shirt	$10,332.00
Dress	$10,320.00
Pants	$10,090.00
Jewelry	$10,010.00

Other useful views to build in the notebook include payment-method frequency, average purchase amount by shipping type, category-level review ratings, and a numeric correlation heatmap. Treat correlation as association rather than causation.

7. Business Questions & Recommendations

Recommendations to test

• **Category merchandising:** Review high-total and high-average categories together before changing inventory or promotional placement.

• **Subscription strategy:** Test member benefits and compare repeat-purchase rate and revenue per customer using a dataset that can reliably identify customers over time.

• **Discount effectiveness:** Compare discounted and non-discounted purchases alongside margin and incremental sales data before deciding whether a discount is profitable.

• **Product experience:** Combine product sales with review ratings; investigate low-rated products before increasing promotion.

• **Customer segmentation:** Define segments using transparent purchase-history rules and validate them against customer-level records. A row-level dataset should not be mistaken for one row per unique customer.

Next steps

1. Complete and validate the Python notebook, saving the cleaned dataset and chart outputs.

2. Rebuild SQL queries against the cleaned dataset and record reproducible outputs.

3. Recreate the Power BI report with clear measures, labels, filters, and documented calculations.

4. Add dashboard screenshots and a README to the GitHub repository.

5. Keep the raw dataset unchanged and document all transformations.

Tools & reproducibility

Python libraries: pandas, numpy, matplotlib, seaborn. Suggested project files: raw CSV, cleaned CSV, Jupyter notebook, SQL script, Power BI PBIX, report PDF, screenshots, README, and requirements.txt.

Source note

Dataset and initial reference scope were supplied by the user in the uploaded project archive. The uploaded reference report credits WeXllent tutorial materials. This newly generated report is a restructured, independently calculated descriptive summary of the supplied CSV; it is not a certification of original dataset ownership or of a separately rebuilt Power BI dashboard.